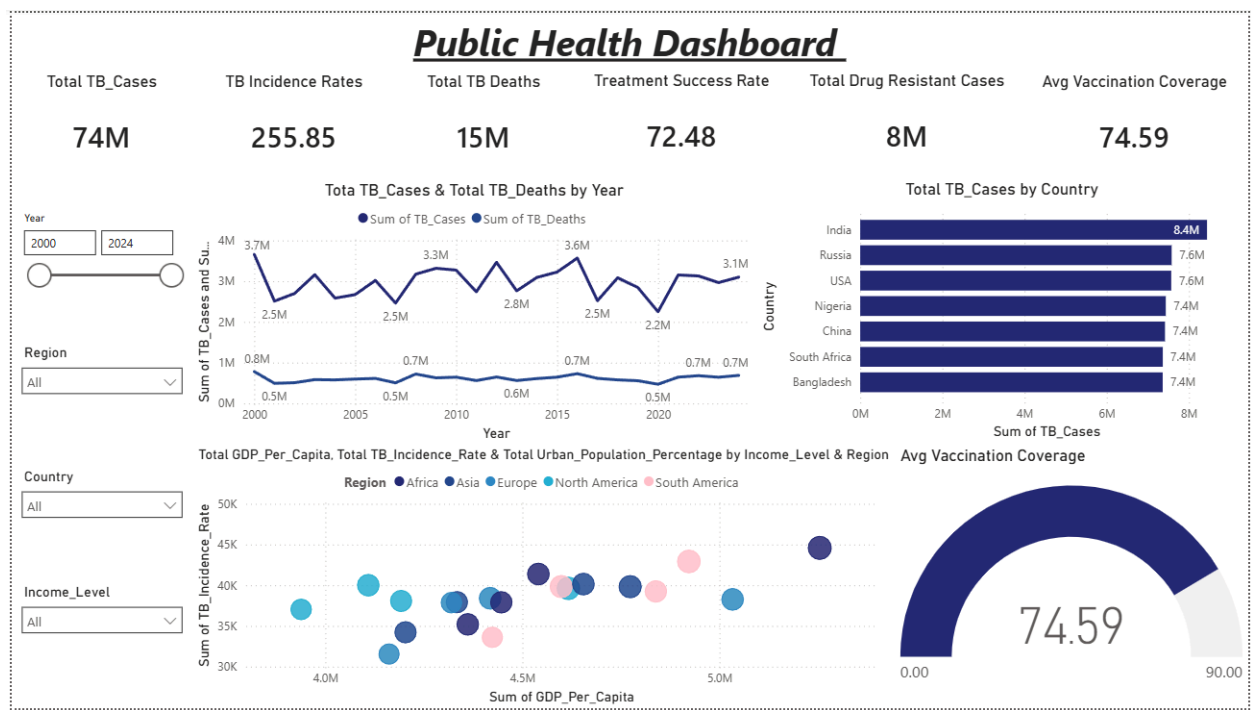


Task 14.5 Public Health Dashboard Report

Overview

The Public Health Dashboard presents a comprehensive analysis of Tuberculosis (TB) trends across multiple countries and regions over time. It integrates epidemiological data with socio-economic and healthcare indicators to evaluate disease burden, treatment outcomes, and prevention efforts.



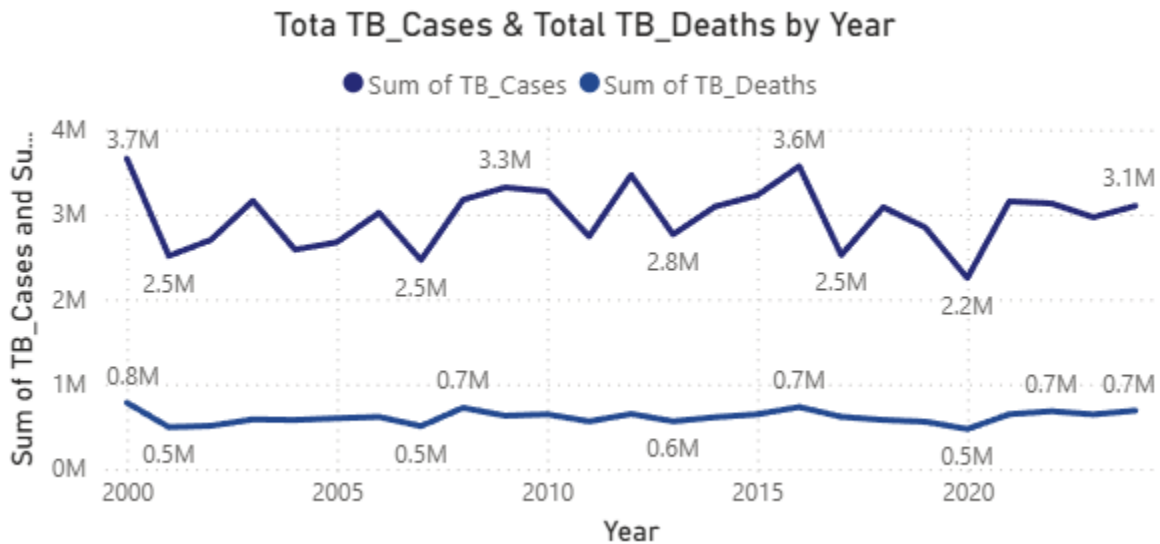
Key Metrics Summary

Total_TB_Cases	TB_Incidence_Rates	Total_TB_Deaths	Treatment_Success_Rate	Total_Drug_Resistant_Cases	Avg_Vaccination_Coverage
74M	255.85	15M	72.48	8M	74.59

- **Total TB Cases:** 74 Million
- **TB Incidence Rate:** 255.85
- **Total TB Deaths:** 15 Million
- **Treatment Success Rate:** 72.48%
- **Total Drug Resistant Cases:** 8 Million

- **Average Vaccination Coverage:** 74.59%
-

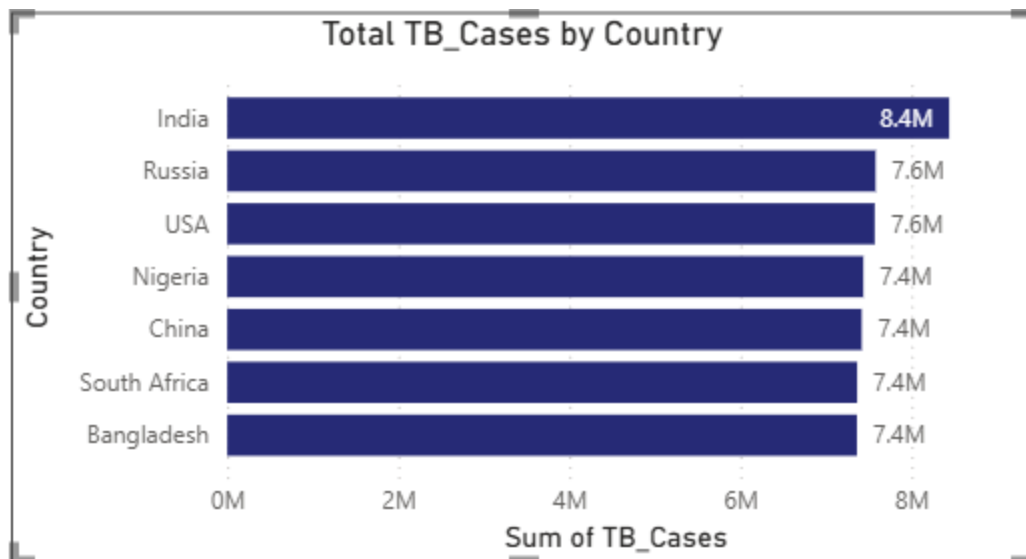
Trend Analysis (2000–2024)



The time-series analysis of TB cases and deaths indicates fluctuations over the years.

- TB cases show periodic increases and decreases, with peaks around mid-2000s and mid-2010s.
 - TB deaths remain relatively stable compared to cases but follow a similar trend pattern.
 - Recent years show a slight stabilization in both cases and deaths.
-

Country-Level Analysis

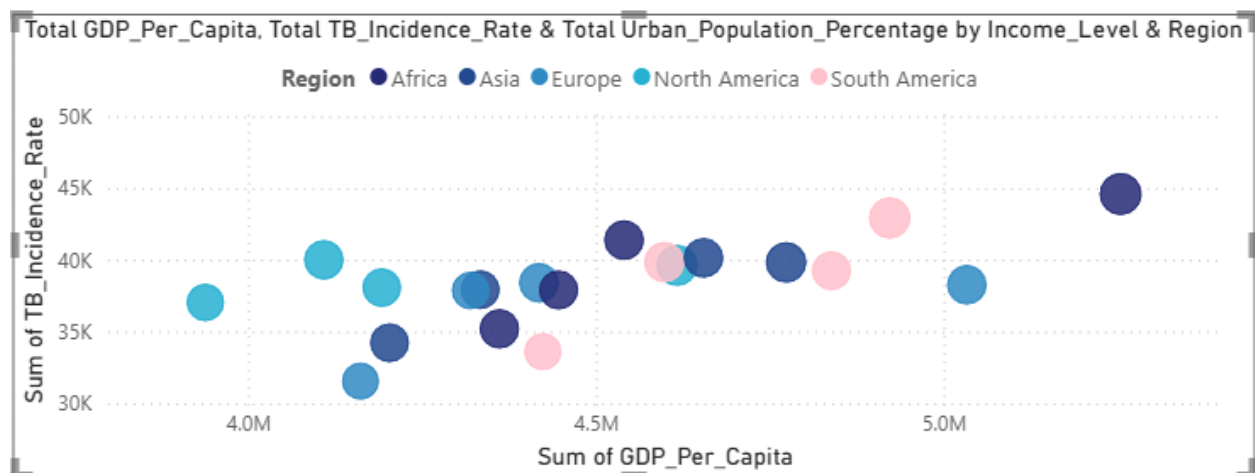


The distribution of TB cases across countries highlights the highest contributors:

- India reports the highest number of TB cases at approximately 8.4 million.
- Russia and the USA follow with around 7.6 million cases each.
- Nigeria, China, South Africa, and Bangladesh also contribute significantly, each reporting around 7.4 million cases.

This indicates a concentration of TB burden in specific high-population countries.

Socio-Economic and Regional Analysis

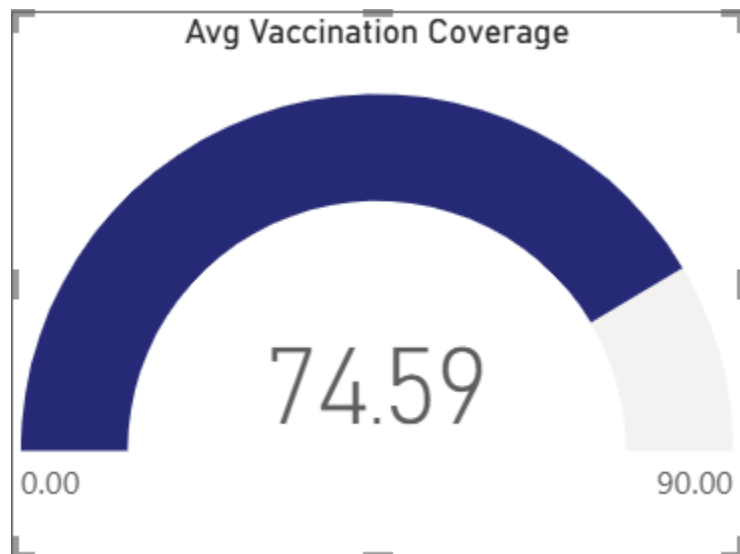


The relationship between economic indicators and TB incidence reveals:

- Countries with varying GDP per capita show differences in TB incidence rates.

- Regions such as Africa, Asia, Europe, North America, and South America display diverse patterns in disease prevalence.
 - Urban population percentage and income level influence TB incidence across regions.
-

Healthcare and Vaccination Insights



- The average vaccination coverage stands at **74.59%**, indicating moderate immunization levels.
 - Treatment success rate is **72.48%**, reflecting ongoing efforts in TB management.
 - Despite these efforts, the presence of **8 million drug-resistant TB cases** highlights a significant public health challenge.
-

Interactive Filters (Slicers)

The dashboard incorporates multiple slicers to enable dynamic data exploration and enhance user interactivity. These filters allow users to analyze TB trends across different dimensions.

Year

Year ▼

A horizontal timeline slider with two circular endpoints connected by a line.

- Allows selection of a specific year or a range of years (2000–2024).
- Enables analysis of temporal trends in TB cases and deaths.

Region

Region

▼

- Filters data by geographical regions such as Asia, Africa, Europe, North America, and South America.
- Supports comparative analysis of TB burden across regions.

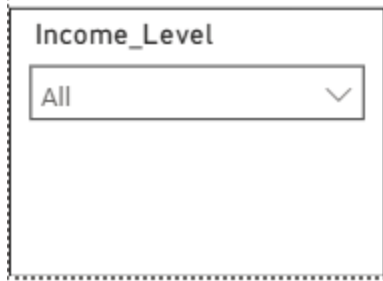
Country

Country

▼

- Enables selection of individual countries.
- Facilitates detailed country-level analysis of TB cases, deaths, and related indicators.

Income Level

A rectangular box with a dashed border. Inside the box, at the top, is the text "Income_Level". Below this text is a dropdown menu. The dropdown menu has a white background and a thin border. It contains the word "All" on the left and a small downward-pointing chevron icon on the right.

- Categorizes countries based on income groups (e.g., Low, Lower-Middle, Upper-Middle, High).
- Helps analyze the relationship between economic status and TB incidence.

Conclusion

The dashboard highlights the global burden of TB, emphasizing variations across countries, regions, and economic conditions. While vaccination and treatment programs show measurable progress, the persistence of high case numbers and drug-resistant TB underscores the need for continued monitoring and intervention.